Muhammad Raheel Anwar

+92-302-9802838 | pingmra@gmail.com | IELTS - 6.5

linkedin.com/in/itxmra | github.com/raheelism | raheelism.github.io

NUST, Islamabad, Pakistan

Objective

Seeking a challenging position to apply my expertise in **Agentic AI, Large Language Models (LLMs), and Vision-Language Models (VLMs)**. My aim is to contribute to innovative projects at the intersection of **intelligent automation, multi-agent systems, and advanced AI architectures**, leveraging deep learning and generative AI to solve complex real-world problems.

Experience

- **Systems Limited x McKinsey & Company** *Nov 2025 – Present*
  Consultant AI Islamabad, Pakistan
  - Providing AI consultancy to McKinsey & Company and developing enterprise-grade solutions leveraging LLMs, RAG architectures, and multimodal AI.
  - Collaborating with cross-functional teams to integrate AI-driven automation and insights into business processes.
  - **Enterprise Chatbots & Automation:** Developing scalable RAG-based chatbots using Azure AI Foundry and LangChain, automating customer support workflows and reducing response times.
  - **MLOps & Infrastructure:** Implementing end-to-end MLOps pipelines on Databricks for model lifecycle management, ensuring reproducible training and deployment via Docker and Kubernetes.
- **Intelligent Robotics Lab, NCAI, NUST** *Mar 2025 – Nov 2025*
  Senior Research Officer Islamabad, Pakistan
  - Led research in Agentic AI and RAG-based systems, including VLM-based detection (FastVLM) and plant disease detection.
  - **Foundational Urdu LLM:** Led the development of a custom Urdu Large Language Model, including tokenizer training, distributed pre-training on high-performance computing clusters, and fine-tuning for downstream tasks.
  - **Autonomous Coding Agent:** Engineered a specialized agentic framework architecture for codebase manipulation, implementing Model Context Protocol (MCP) for seamless tool integration and designing multi-agent architectures that automate refactoring and feature implementation.
  - **AI-Code Security & Test Automation:** Developed an automated verification pipeline for AI-generated code, utilizing static and dynamic analysis to detect vulnerabilities and ensuring production-readiness through rigorous security-focused test generation.
  - **Juntendo University Medical System:** Architected a comprehensive medical advice system integrating VLM-based diagnostics and LLM-powered patient interaction, deployed as a full-stack mobile solution.
  - **FastVLM & Plant Disease Detection:** Pioneered research in VLM localization (FastVLM) and developed a plant disease detection system achieving 28% higher accuracy than traditional CNNs.
  - **Custom Voice Agents:** Developed real-time, low-latency conversational voice agents for university help desks, integrating GPT-4, ElevenLabs, and VAPI.
- **Aqsa Bajwa SMC Ltd** *Jan 2024 – Mar 2025*
  AI Consultant Islamabad, Pakistan
  - **Agentic Content Generation:** Built an autonomous blog post generation agent using Model Context Protocol (MCP) and WordPress integration, automating content creation and SEO optimization.
  - **Generative Fashion Design:** Developed a dress design system using fine-tuned Diffusion models, coupled with an automated product upload pipeline that generates titles, descriptions, and SEO tags.
  - **Inventory & Sales Intelligence:** Implemented predictive analytics for sales forecasting and inventory optimization, utilizing advanced regression models to drive data-backed business decisions.
  - **ResumeParser:** Built an intelligent resume extractor using Spacy, OCR, and Llama models to automate candidate screening.
- **Sparkleo Tech** *Apr 2024 – Sept 2024*
  Mobile Development with AI Integration Islamabad, Pakistan
  - **AI-Integrated Mobile Ecosystems:** Integrated advanced AI modules into mobile applications, including next-word/sentence prediction engines, generative icon creation, and real-time voice recognition.
  - **Smart AAC Solutions:** Developed predictive communication systems for Augmentative and Alternative Communication (AAC) tools, significantly improving accessibility for users with disabilities.
  - **Adaptive Teaching Assistant:** Engineered a multi-agent educational assistant using LangChain and GPT-4o, featuring interactive quizzes and personalized learning paths.

- **IZBA - CAPABL** *Aug 2022 – Dec 2024*
  Researcher USA - Remote
  - **Autonomous Data Agents:** Deployed intelligent scraping and verification agents to autonomously aggregate and validate logistics data from diverse web sources.
  - **Supply Chain Predictive Modeling:** Developed robust machine learning models to predict missing values and forecast trends for 3PL and 4PL logistics providers, enhancing data integrity and operational planning.
  - Conducted extensive data preprocessing and analysis to support machine learning pipelines.
- **Qureshi Creatives** *Apr 2021 – Mar 2022*
  Intelligent Web Application Developer Islamabad - Remote
  - **Intelligent Recommendation Engine:** Built a sophisticated web-based recommendation system for Lego toys and BrickLink products, utilizing collaborative filtering to personalize user experiences.
  - Implemented ML features in cross-platform web solutions with optimized performance and Firebase integration.
- **NFC IET University** *Sept 2020 – May 2021*
  Lecturer Multan, Pakistan
  - Taught Mobile App Development and Game Development.
  - Mentored final-year projects in AR and interactive applications.
- **Xufire Tech** *Dec 2020 – Apr 2021*
  Software Engineer Multan, Pakistan
  - Engineered and deployed on-device ML models for mobile applications focusing on real-time inference and optimization for resource-constrained environments.
- **Infotech4IT** *Jun 2018 – Jul 2020*
  Software Engineer Lahore, Pakistan
  - Developed data-driven UI components and integrated RESTful APIs to serve predictive model outputs within enterprise-grade analytics dashboards.
  - Optimized data ingestion layers for ML training pipelines, ensuring high data integrity and performance during large-scale model evaluation cycles.

Education

- **National University of Sciences & Technology (NUST)** *Oct 2022 – July 2025*
  *M.Sc. in Computer Science - **GPA**: 3.70/4.00* Islamabad, Pakistan
  - Thesis: *MCQs Generation Using Large Language Models: A Mixture-of-Agent Approach*
  - Relevant Coursework: Generative AI, Deep Learning, Machine Learning, Blockchain Technologies
- **NFC Institute of Engineering & Technology (NFC IET)** *Oct 2016 – Sept 2020*
  *B.Sc. in Computer Science - **GPA**: 3.49/4.00* Multan, Pakistan
  - Projects: *Believers Game* and *Dynamic Shopping App*
  - Relevant Coursework: Object-Oriented Programming, Algorithms, Operating Systems

Patents and Publications

C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis

[S.2] Raheel Anwar et al. (2025). **MCQs Generation Using Large Language Models: A Review**. Manuscript accepted for publication in *IEEE Access*. Published: January 16, 2026. https://doi.org/10.1109/ACCESS.2026.3652006

[S.2] Raheel Anwar et al. (2025). **MCQsAgent: Towards Human-Level MCQs Generation via Collaborative Multi-Agent AI Framework**. Manuscript accepted for publication in *Expert Systems With Applications*. Accepted: January 30, 2026.

[S.3] Raheel Anwar et al. (2026). **MSRANet: A Multiscale Residual Attention UNet for Brain Tumor Segmentation and Classification**. Manuscript under review for publication in *IEEE Transactions on Medical Imaging*. Submitted: December 9, 2025.

[S.4] Raheel Anwar et al. (2025). **Towards Agentic AI: A Systematic Review of LLM-based Multi-Agent Intelligent Systems**. Manuscript under review for publication in *Artificial Intelligence Review*. Submitted: September 11, 2025.

[S.5] Raheel Anwar et al. (2025). **Plant Disease Detection Using Vision Language Models: A Comparative Study**. Manuscript submitted to *Precision Agriculture*. Under Review: July 7, 2025.

[S.6] Raheel Anwar et al. (2026). **When Does Chain-of-Thought Hurt Reranking? A Systematic Zero-Shot Analysis**. Manuscript under final stages

[T.1] Raheel Anwar et al. (2025). **Improving MCQs Generation Using Large Language Models: A Mixture of Agents Approach**. Thesis.

Skills

- **AI & Machine Learning:** Agentic AI, LLMs, RAG, Finetuning, Deep Learning, Computer Vision, NLP, Multi-Agent Systems
- **Languages & Frameworks:** Python, TensorFlow, PyTorch, LangChain, OpenCV, Scikit-learn, C++, Dart, Java
- **MLOps & Cloud:** MLflow, Docker, Kubernetes, AWS, Azure, Databricks, AI Foundry, CI/CD
- **Web & Mobile:** FastAPI, Flask, Flutter (Integration), REST APIs
- **Databases:** PostgreSQL, MySQL, MongoDB, VectorDB, ElasticSearch
- **Research Skills:** Systematic Literature Review, Academic Writing, Experimental Design, Data Analysis

Awards and Recognitions

- **Nominated for Best Methodology Award - MS Thesis** *2025*
  Graduate Studies Committee
  ◦ Nominated for the Best Methodology Award for innovative approach in MS thesis research.
- **Entrepreneurial Marketing Course** *2023*
  Online Program
  ◦ Delivered a 4-week online course focused on marketing strategies for startups and entrepreneurs.
- **"How to Start a Business" Online Session** *2023*
  Online Event
  ◦ Hosted a live online session for over 100 participants on starting a business from scratch.
- **Guest of Honor - Freelancers Top Voice Event** *2022*
  International Islamic University (IIUI), Islamabad
  ◦ Invited as Guest of Honor at the "Freelancers Top Voice" event organized by IIUI Islamabad.
- **"How to Build a Startup" Introductory Session** *2022*
  NANSUA University, Multan
  ◦ Led a 4-week session introducing students to startup ideation, business models, and pitch preparation.
- **Freelancing Training Session for Women University, Multan** *2021*
  Women University, Multan
  ◦ Delivered a 4-week freelancing training session aimed at empowering female students with remote work skills.
- **AI Talk Seminar** *2020*
  NFC Institute of Engineering & Technology (NFCIET), Multan
  ◦ Delivered a seminar on the fundamentals and future of Artificial Intelligence.
- **Best Duo Final Year Project - Dynamic Adaptive UI App** *2020*
  NFC Institute of Engineering & Technology (NFCIET), Multan
  ◦ Developed a dynamic and adaptive UI mobile app that can be converted into multiple domains like grocery, pharmacy, etc.
- **Finalist Winner, ICE 2019 - Deev: Ride Sharing App** *2019*
  Innovation and Creativity Event (ICE), NFCIET Multan
  ◦ Finalist winner with "Deev", a ride-sharing mobile app built in Flutter.
- **Runner-up, ICE 2018 - Anonymous Web Browser** *2018*
  Innovation and Creativity Event (ICE), NFCIET Multan
  ◦ Developed an anonymous and lightweight web browser as part of the Innovation and Creativity Event.

Leadership Experience

- **Co-Founder & CEO** *Sept 2024 - Present*
  Aqsa Bajwa Designer Clothing Brand, Islamabad [🌐]
  ◦ Launched Aqsa Bajwa, a designer clothing brand with exclusive fashion collections.
  ◦ Achieved PKR 1 million in sales within the first month of operations.
- **Founder & Product Lead** *Jan 2022 - Present*
  InShort Tech (Edutech Startup), Islamabad
  ◦ Founded InShort Tech, developing educational apps such as Coursely, Scholarly, and InShort Quiz.
  ◦ Delivered over 20+ mobile development projects for clients across education and productivity domains.
- **Founder & Brand Strategist** *Apr 2021 - Present*
  Barcurs Premium Sunglasses, Islamabad [🌐]
  ◦ Created Barcurs, a premium sunglasses brand targeting international markets including the US and Europe.
- **Founder & E-commerce Developer** *2020 - 2021*
  BlackWears Clothing Brand, Multan
  ◦ Built BlackWears, a drop-shipping-based clothing store focused on casual wear and accessories.
  ◦ Designed and managed an online store using Shopify, handling product sourcing and fulfillment.

Volunteer Experience

- **Smart Research Workshop: Tools & Strategies for High-Impact Publications** *Aug 2025*
  Intelligent Robotics Lab (IRL), NCAI, Pakistan [🌐]
  - Delivered an AI-driven research workshop on literature review, reference management, and paper structuring, attended by students and early-career researchers nationwide.
- **Community Tech Mentor** *Jun 2021 - Aug 2021*
  Punjab IT Board, Lahore, Pakistan [🌐]
  - Mentored under the Punjab Youth Internship Program, conducting workshops on mobile app development and freelancing basics for youth.
- **Online Education Volunteer** *Apr 2020 - Jun 2020*
  Saylani Welfare Trust, Karachi, Pakistan [🌐]
  - Tutored students in computer science fundamentals during the pandemic, supporting Saylani's online education initiative.
- **Event Coordinator** *Feb 2019 - Mar 2019*
  National Science and Technology Park (NSTP), Multan [🌐]
  - Coordinated a regional tech innovation fair, managing logistics and presentations for 30+ student startup teams.

Certifications

- **Credly Certifications (12 Total)**
- **Game Design and Development Specialization – Michigan State University**
- **Machine Learning – Stanford University**
- **Game Development for Modern Platforms – Michigan State University**
- **What is Data Science? – IBM**
- **Data Science Methodology – IBM**
- **Game Design and Development Capstone – Michigan State University**
- **Microsoft Azure Fundamentals (AZ-900) – Microsoft Press and Jim Cheshire**
- **Building RAG Solutions with Azure AI Foundry – Ziggy Z**

Additional Information

Languages: English (Fluent), Urdu (Native), Punjabi (Intermediate), Arabic (Beginner)
Interests: Gaming, Singing, Rapping, History, Exploring new tools in tech

References

1. **Dr. Shah Khalid**
   Assistant Professor, Department of Computing
   School of Electrical Engineering and Computer Science (SEECS)
   National University of Sciences and Technology
   H – 12 Islamabad, Pakistan
   Email: shah.khalid@seecs.edu.pk
   Relationship: Thesis Advisor
2. **Hafiz Syed Muhammad Bilal**
   Assistant Professor, Department of Computing
   School of Electrical Engineering and Computer Science (SEECS)
   National University of Sciences and Technology (NUST)
   H – 12 Islamabad, Pakistan
   Email: bilal.ali@seecs.edu.pk
   Relationship: GC Member